



AMD TCS Hackathon Handbook

AIG Team
June 6, 2026

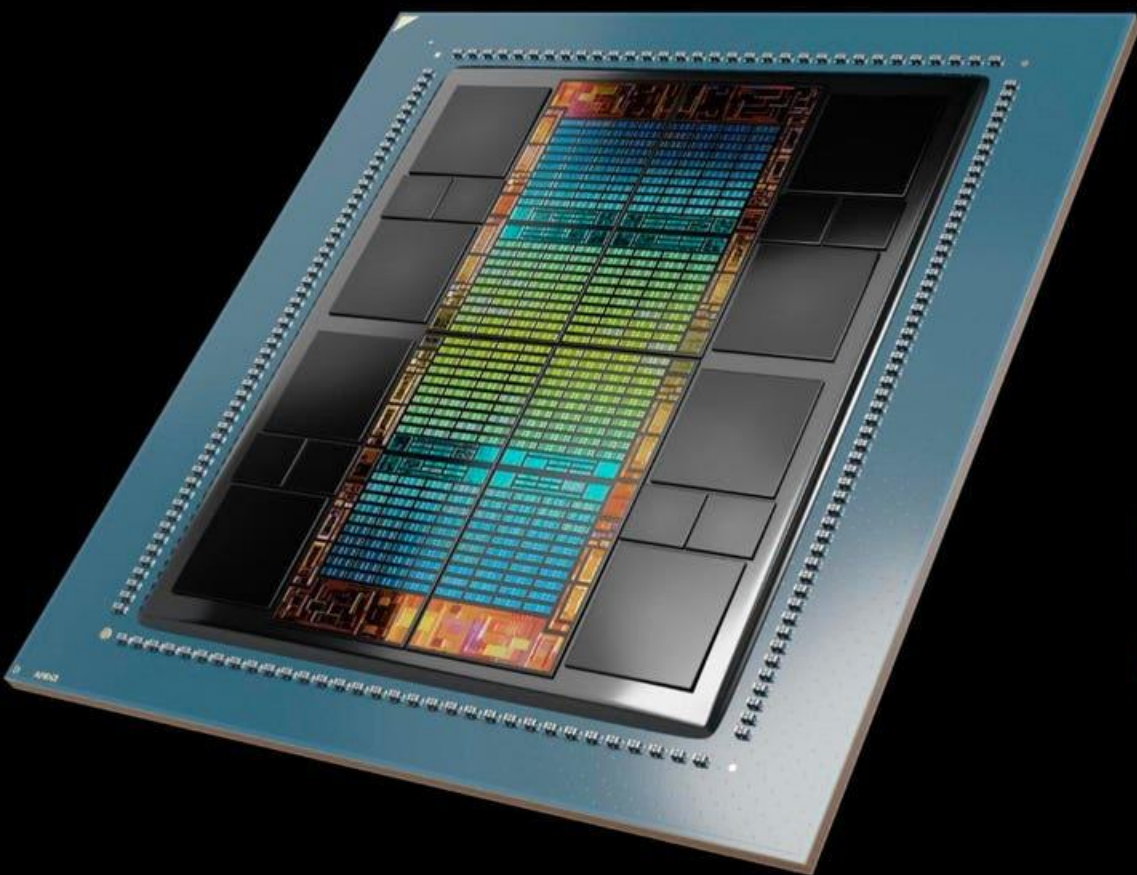
AMD 
together we advance_

An opportunity to build AI
workloads on powerful AMD GPUs
in a cloud environment.
No hardware setup required.



Agenda

1. **AMD GPU – Instinct MI300X**
2. **ROCm Software Stack**
3. **Development Environment**
4. **Submission**



AMD Instinct™ MI300X

Leadership generative AI
accelerator

AMD
CDNA 3

192 GB
HBM3

~5.3 TB/s
Memory Bandwidth
(Peak Theoretical)

Up to 896 GB/s
AMD Infinity Fabric™ Bandwidth

Memory Capacity and Performance Characteristics

Feature	MI300X Specification	Relevance to AI
HBM Capacity	192 GB HBM3	Enables very large models on a single GPU
Memory Bandwidth	~5.3 TB/s	Reduces memory bottlenecks in transformers
Precision Support	FP16, BF16, FP8	Optimized for training and inference

MI300X Excels At

Large Language Model Training

Supports training massive models with 192 GB HBM, reducing parallelization complexity and device count.

Generative AI Inference

Efficient single-GPU hosting of chatbots and content generation improves deployment and lowers latency and costs.

High Performance Computing

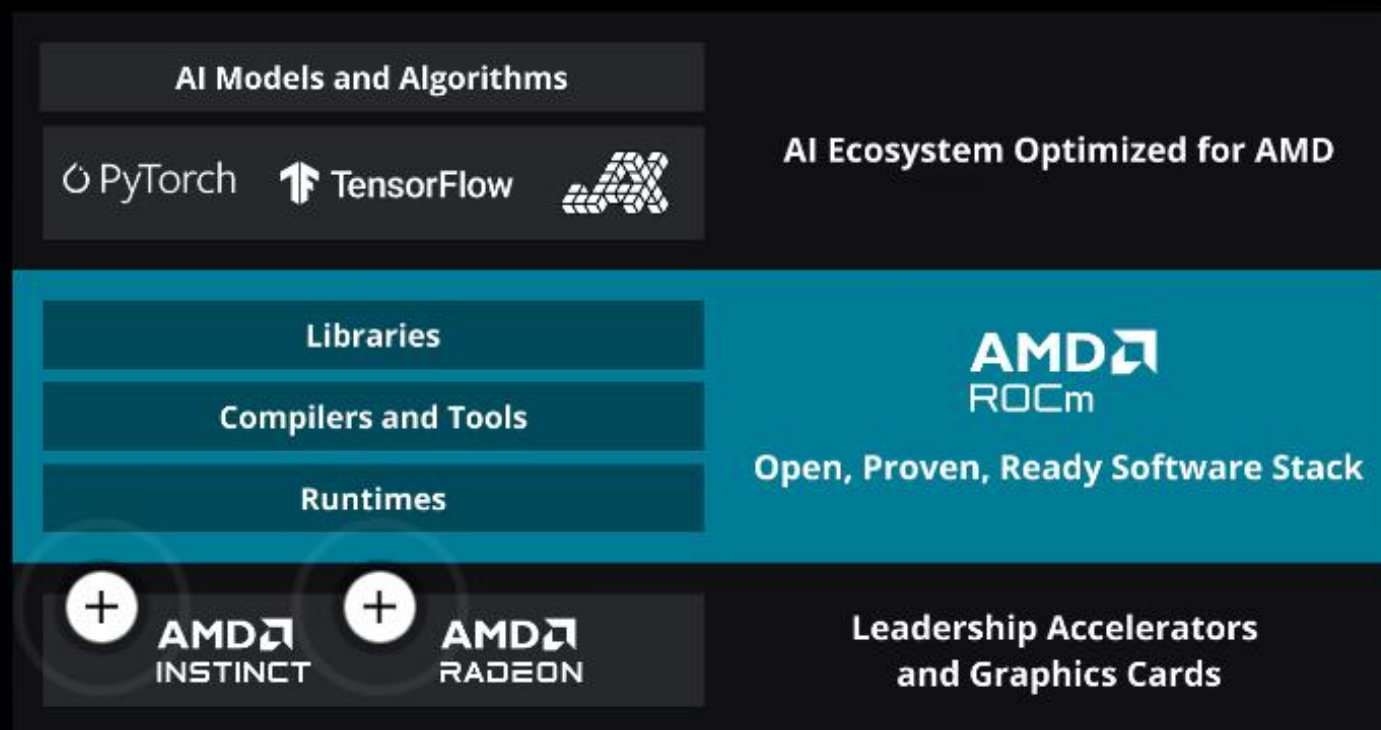
MI300X excels in HPC tasks like fluid dynamics and climate modeling with high bandwidth and compute precision.

System Simplicity and Scalability

Reducing accelerator count lowers network complexity, power use, and software overhead for scalable systems.

ROCm Software

Open software stack that includes programming models, drivers, tools, compilers, libraries, APIs and runtimes that enable GPU programming from low-level kernel to end-user applications. ROCm is optimized for Generative AI and HPC applications, and it is easy to migrate existing code into ROCm software.



What's New in ROCm 7

Full support for AMD Instinct™ MI350 Series GPUs

Distributed inference with open-source framework support

Enterprise-ready AI tools with orchestration and endpoint integration

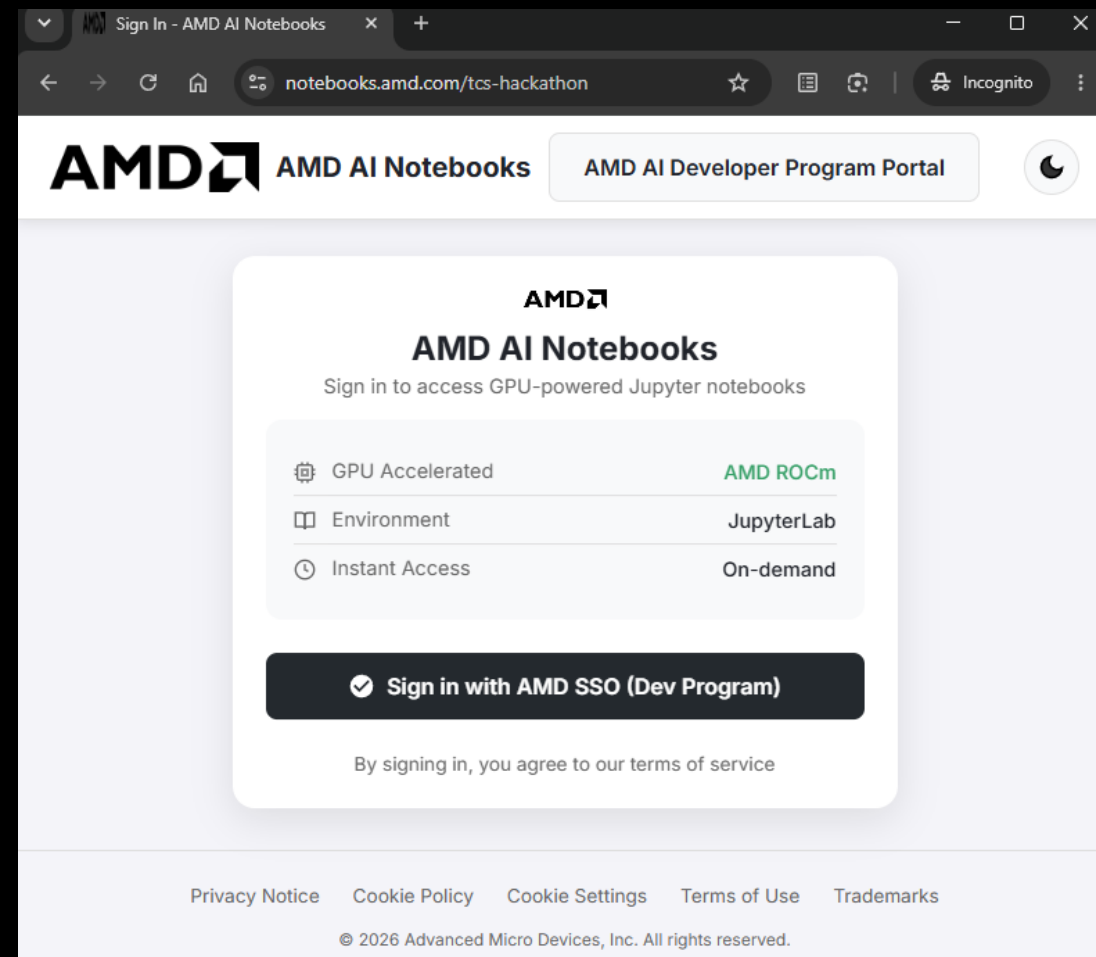
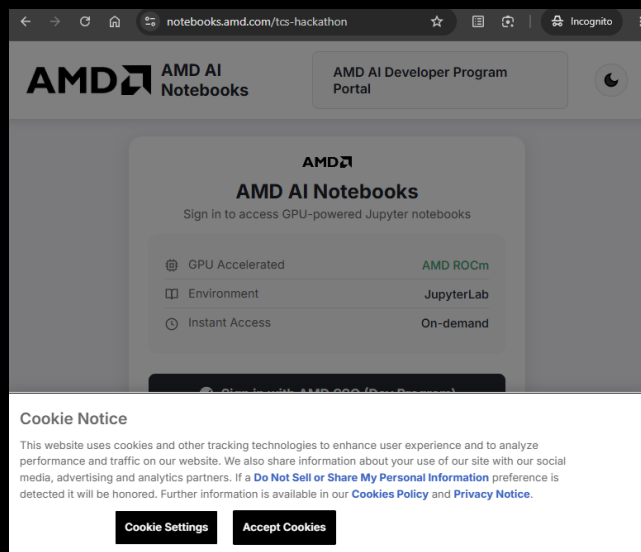
Support for large-scale models with new data types FP6 and FP4 for enhanced flexibility and performance.

Enhanced code portability with HIP 7.0

<https://rocm.docs.amd.com/en/latest/about/release-notes.html>

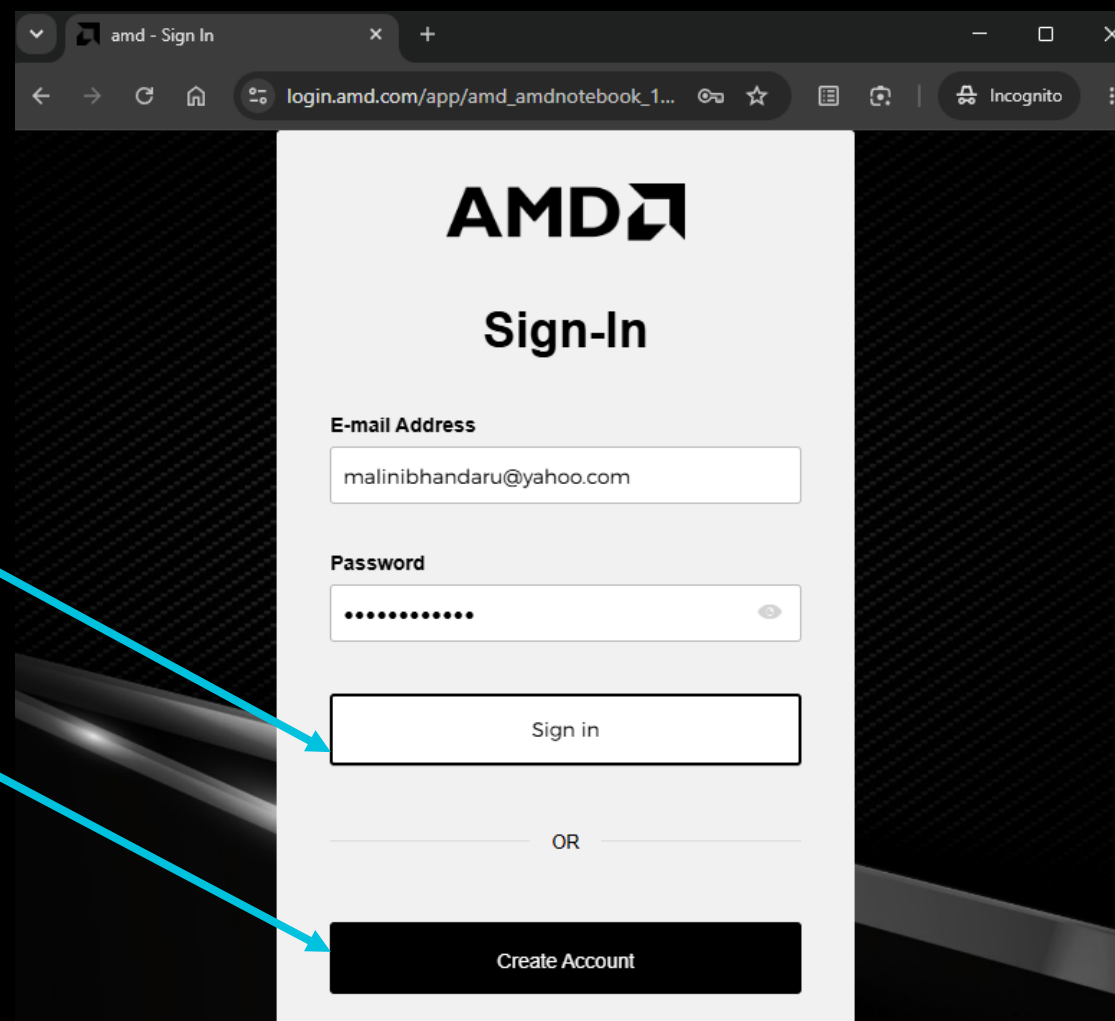
AMD Development Environment

<https://notebooks.amd.com/tcs-hackathon>



Sign-into or Create Your Account

If you have an AMD AI Dev Program Account, use it, else sign up using your TCS email ID



The screenshot shows a web browser window with the title 'amd - Sign In'. The address bar displays 'login.amd.com/app/amd_amdnotebook_1...'. The page features the AMD logo at the top, followed by the heading 'Sign-In'. Below this, there are two input fields: 'E-mail Address' with the value 'malinibhandaru@yahoo.com' and 'Password' with masked characters. A 'Sign in' button is positioned below the password field. A horizontal line with the text 'OR' is centered below the 'Sign in' button. At the bottom, there is a dark blue button labeled 'Create Account'. Two red arrows originate from the text box on the left: one points to the 'Sign in' button and the other points to the 'Create Account' button.

Creating your AMD AI Dev Program Account

Sign up using your
TCS email ID

The screenshot shows a web browser window with the URL `amd.com/en/registration/ai-dev-program-sig...`. The page title is "AMD AI Developer Program Sign Up Form".

AMD AI Developer Program Sign Up Form

Thank you for your interest in the AMD AI Developer Program.

By joining this program, you will receive periodic program emails notifying you about member benefits, upcoming events, new technical resources, and sweepstakes announcements.

Start Accessing Your Member Benefits

Step 1: Create your **free** AMD account and complete the AI Developer Program Sign Up Form.

Step 2: Verify your email. An account activation message with an Access Token will be sent via e-mail to the address you specify on the form.

Step 3: After verification, you will be redirected to access the AMD AI Developer Program Portal.

Have questions or need help signing up? Reach out to us at ai_dev_program@amd.com

Form Fields:

- First Name *
- Last Name *
- E-mail *
- Company / University Name *
- Please enter your company or university name, or "None" if not affiliated.
- Location *
Select one
- Preferred Language *
English
- Where will you primarily deploy and run your projects?
☐ Cloud/data center ☐ Laptop, tablet, or mini PC ☐ Workstation or desktop
☐ Edge devices
- ☐ By checking this box and submitting my information to join the AMD Developer Program, I agree to be automatically entered into periodic sweepstakes promotions sponsored by AMD for a chance to win prizes. Open only to legal residents of the United States (DC) and Canada who are 18 years of age or older at the time of entry. Void where prohibited. NO PURCHASE NECESSARY. Odds of winning depend on the number of eligible entries received. See [Official Rules](#) for complete details.
- ☐ I also want to receive technical and marketing communications from AMD like software release notifications and product newsletters. You can opt out at any time [here](#).

Read about how we handle your personal data, your personal data rights, and how you can contact us in our [privacy notice](#).

By creating an account, you agree to the AMD [Terms of Use](#) and [Privacy Policy](#).

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Send a verification email to **m***u@yahoo.com** by clicking on "Send me an email".

Send me an email

Help

Terms of Use

Privacy





Verify with your email

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We sent an email to **m***u@yahoo.com**. Enter the verification code in the text box.

 Haven't received an email? [Send again](#)

Enter Code

614606

Verify

Help

Terms of Use

Privacy

Launch Your Jupyter Notebook Session!

AMD

Welcome back!

Launch your GPU-powered Jupyter notebook

mabhanda

Team: team-3904

Default (ROCm + vLLM)

Request Notebook

?

4 hrs 0 min in 24 hrs

[Sign out](#)

Time available to work



Team Resources – A Pod

10 -13 CPU cores,

240 GB RAM

25 GB of persistent storage

(enough for a large training data set, and
one or two LORA models depending on their size)

16 GB temporary memory

Exclusive access to an AMD MI300X GPU for the duration of your session

192 GB VRAM – enough for a large model (70B comfortable) with KV cache

**Pod may get killed with OOM (out of memory) if
it tries to use excess of 240 GB RAM.**

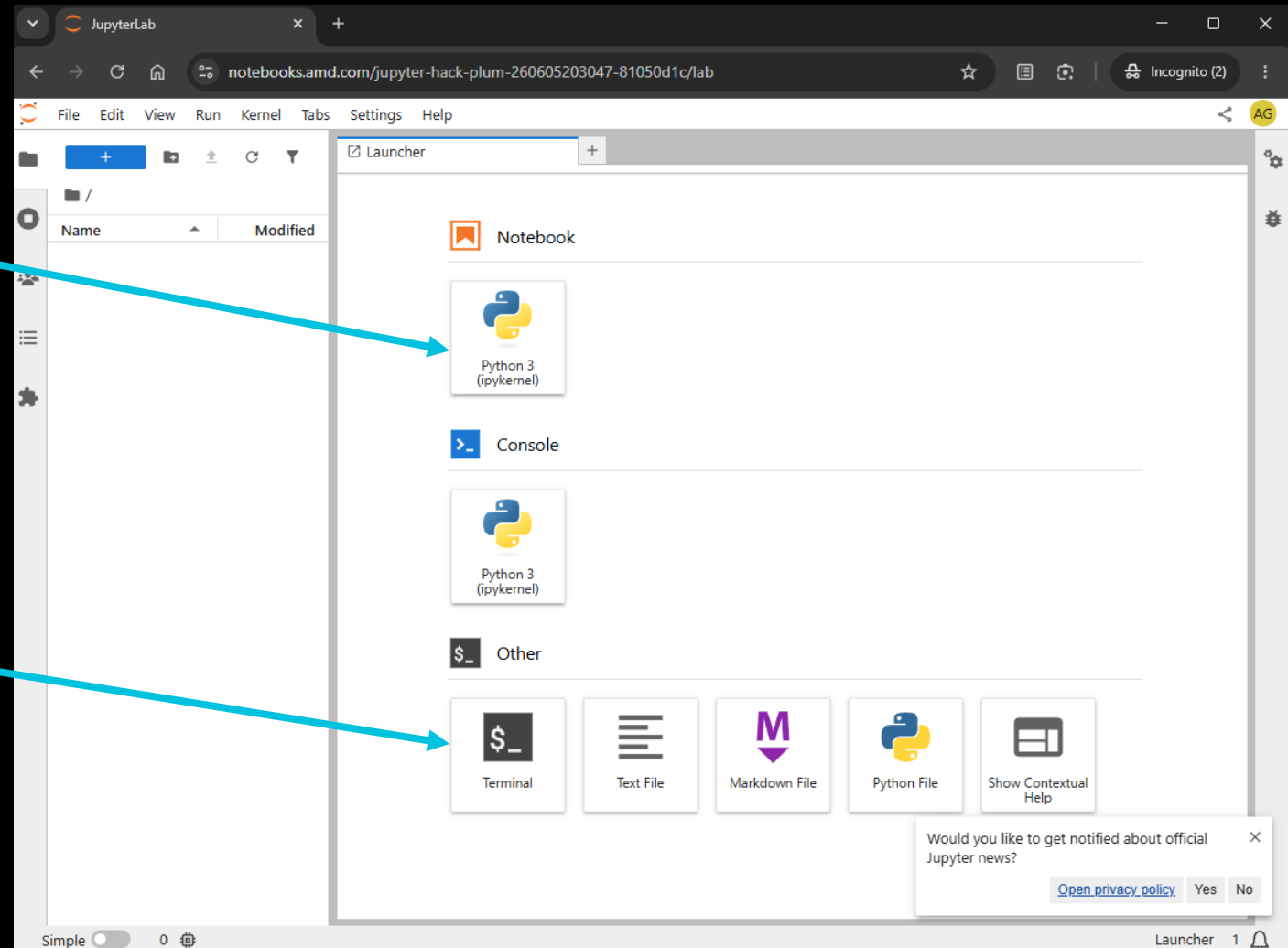
Inside Your Jupyter Notebook Environment

<https://www.youtube.com/watch?v=2WL-XTI2QYI>

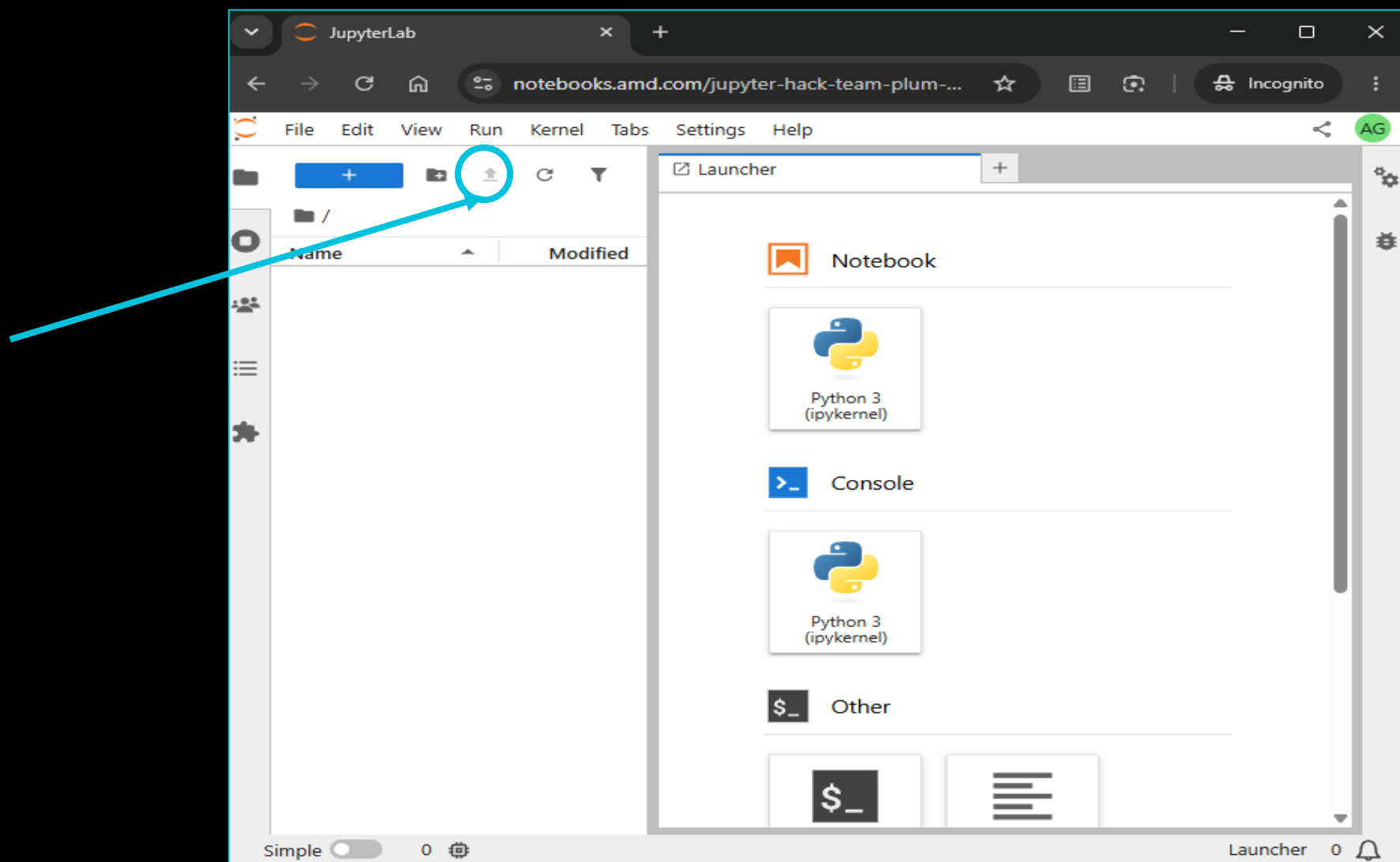
Choose your python kernel

Can start an xterm

- whoami
- amd-smi (shows GPU stats)
- pip install ..
- Serve a model ...



No Uploads from your local to your Jupyter Notebook Environment



Typical Development – Download files, install software, run code

Download a file

```
wget https://huggingface.co/datasets/huggingface/documentation-images/resolve/main/p-blog/candy.JPG
```

Pip install a library

```
pip install -q pydantic_ai openai
```

Clone a repository

```
git clone https://github.com/matrived/blr\_workshop.git
```

Serve a model

```
VLLM_USE_TRITON_FLASH_ATTN=0 vllm serve Qwen/Qwen3-4B --served-model-name Qwen3-4B --api-key abc-123 --port 8000 --enable-auto-tool-choice --tool-call-parser Hermes --trust-remote-code --max_model_len 24272
```

You can “pip install” any software

The screenshot shows a JupyterLab interface in a web browser. The terminal window displays the following content:

```

root@jupyter-hack-team-plum-260605034922-880f8eb7:/workspace#
root@jupyter-hack-team-plum-260605034922-880f8eb7:/workspace#
root@jupyter-hack-team-plum-260605034922-880f8eb7:/workspace# whoami
root
root@jupyter-hack-team-plum-260605034922-880f8eb7:/workspace# amd-smi
+-----+
| AMD-SMI 26.0.0+37d158ab      amdgpu version: 6.16.13  ROCm version: 7.0.0 |
| Platform: Linux Baremetal    |
+-----+
| BDF      GPU  HIP-ID  OAM-ID  Partition-Mode  Mem-Util  Temp  UEC      Power-Usage |
| GPU      |  |  |  |  |  |  |  |  |  |
| 0000:1b:00.0 ...Instinct MI300X OAM  N/A      N/A    0      N/A/750 W |
| 0         0         1      SPX/NPS1  N/A      N/A    0      283/196592 MB |
+-----+
+-----+
| Processes: |
| GPU        PID  Process Name      GTT_MEM  VRAM_MEM  MEM_USAGE  CU % |
+-----+
| No running processes found |
+-----+
root@jupyter-hack-team-plum-260605034922-880f8eb7:/workspace# pip install unsloth
Collecting unsloth
  Downloading unsloth-2026.6.1-py3-none-any.whl.metadata (58 kB)
Collecting unsloth_zoo>=2026.6.1 (from unsloth)
  Downloading unsloth_zoo-2026.6.1-py3-none-any.whl.metadata (33 kB)
Requirement already satisfied: wheel>=0.42.0 in /usr/local/lib/python3.12/dist-packages (from unsloth) (0.45.1)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from unsloth) (25.0)
Requirement already satisfied: torch<2.11.0,>=2.4.0 in /usr/local/lib/python3.12/dist-packages (from unsloth) (2.8.0+gitb2fb688)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (from unsloth) (0.23.0a0+824e8c8)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from unsloth) (2.2.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from unsloth) (4.67.1)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from unsloth) (7.1.0)
Collecting tyro (from unsloth)
  Downloading tyro-1.0.13-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from unsloth) (6.32.1)
Collecting xformers>=0.0.27.post2 (from unsloth)
  Downloading xformers-0.0.35-py39-none-manylinux_2_28_x86_64.whl.metadata (1.2 kB)
Collecting bitsandbytes!=0.46.0,!=0.48.0,>=0.45.5 (from unsloth)
  Downloading bitsandbytes-0.49.2-py3-none-manylinux_2_24_x86_64.whl.metadata (10 kB)

```

The command `pip install unsloth` is circled in green in the original image.

Executing Jupyter Cells

The screenshot shows a Jupyter Notebook in a web browser. The left sidebar displays a file explorer for the workspace `blr_workshop`, listing files like `qwen_image`, `unsloth-finetuning`, `build_airbnb_agen...` (selected), and `PRESENTATION_S...`. The main area contains the following content:

copy & paste this command to launch your vLLM server:

```
VLLM_USE_TRITON_FLASH_ATTN=0 \
vllm serve Qwen/Qwen3-4B \
  --served-model-name Qwen3-4B \
  --api-key abc-123 \
  --port 8000 \
  --enable-auto-tool-choice \
  --tool-call-parser hermes \
  --trust-remote-code \
  --max_model_len 24272
```

Please note that setting of `max_model_len` is to reduce GPU consumption for workshop setting. AMD MI300 GPU can handle much larger `--max_model_len` than the one used for this notebook. This reduced `--max_model_len` is just for the tutorial.

Optionally you can open another terminal and monitor the GPU utilization by running this command:

```
watch rocm-smi
```

After a successful launch, your server should be accepting incoming traffic through an OpenAI-compatible API. Now set some environment variables for your server to use throughout this tutorial:

```
[1]: import os

BASE_URL = f"http://localhost:8000/v1"

os.environ["BASE_URL"] = BASE_URL
os.environ["OPENAI_API_KEY"] = "abc-123"

print("Config set:", BASE_URL)
```

Config set: `http://localhost:8000/v1`

Verify the model is available at the `BASE_URL` you just set by running the following command.

```
[ ]: !curl http://localhost:8000/v1/models -H "Authorization: Bearer $OPENAI_API_KEY"
```

Congratulations, you just launched a powerful server that can serve any incoming request, allowing you to build amazing applications.

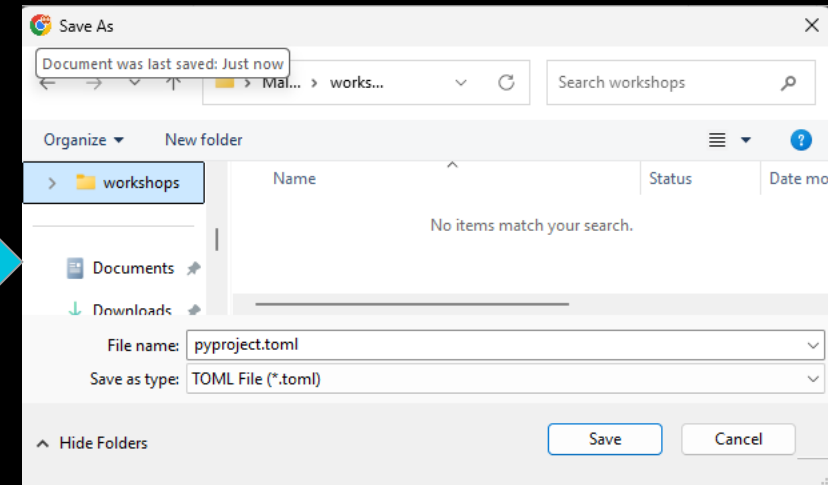
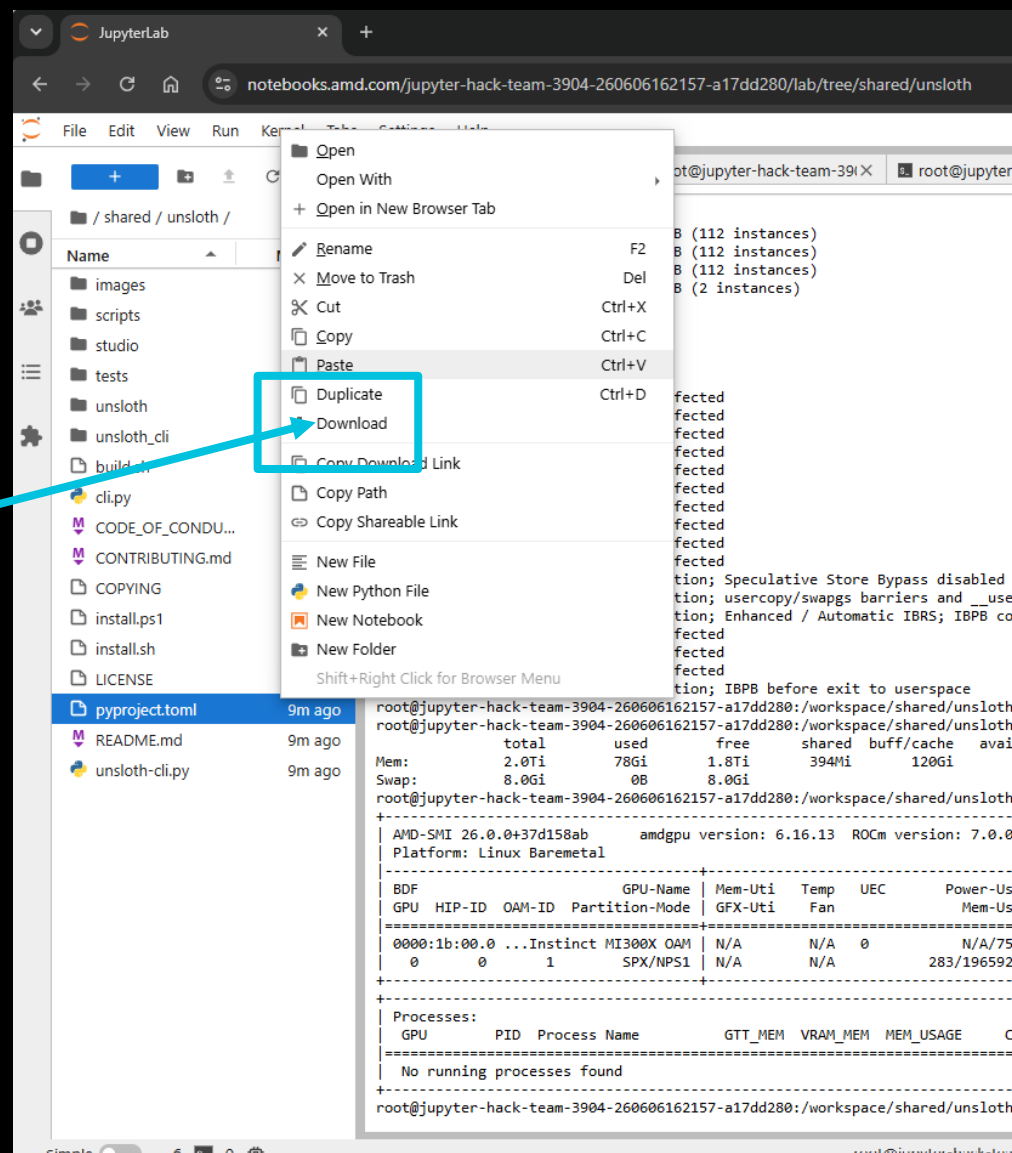
Step 2: Installing dependencies

Install the PydanticAI dependencies using this command:

```
[ ]: !pip install -q pydantic_ai openai
```

Save Files by Downloading to Local Environment

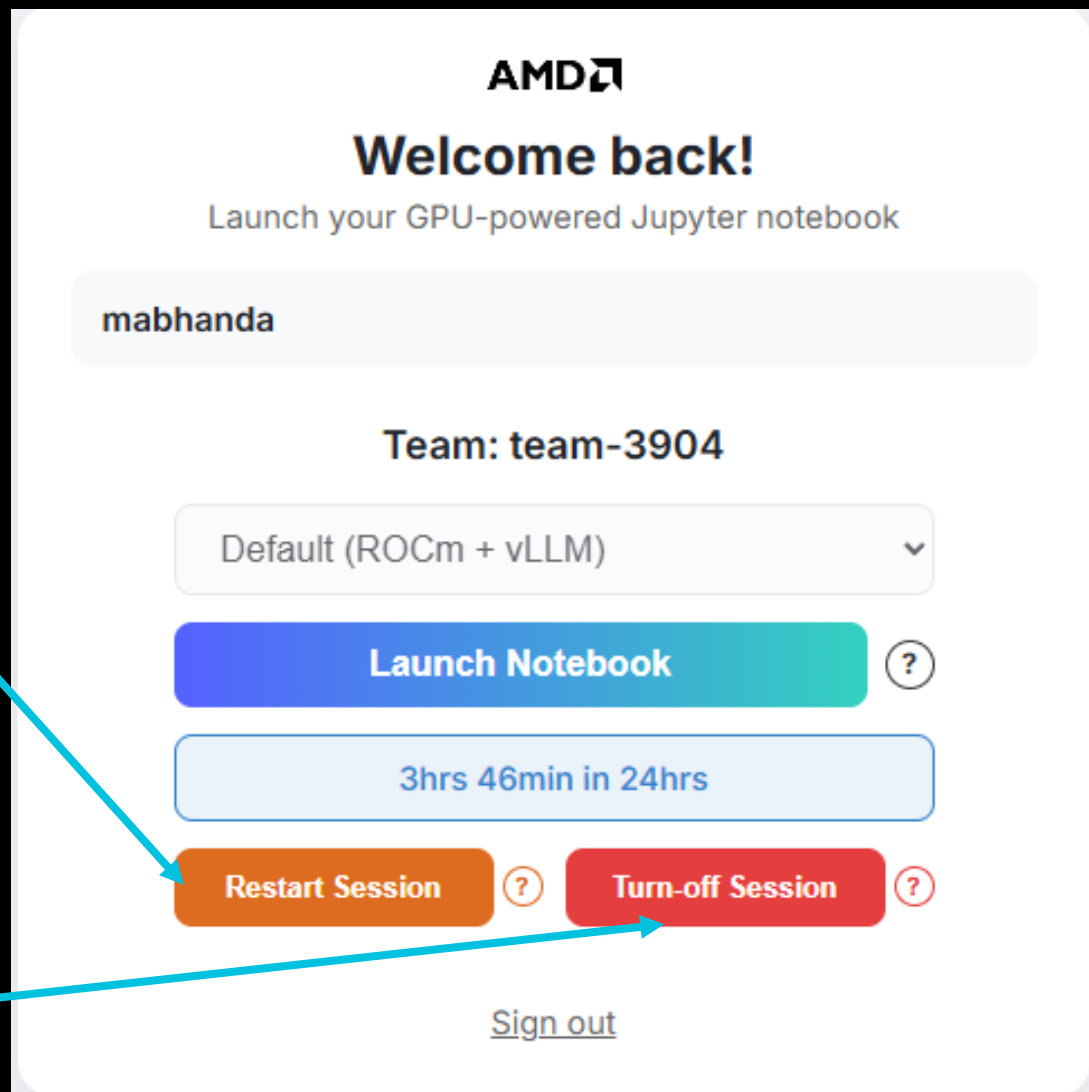
Right click on the file you want to download



Reconnect to a Running Session

Restart Session: Out of Memory (OOM) or Environment Failure, restart with a fresh container. You remaining time carries over.

Turn-off Session: Done with work for a bit. Conserve your time resources and pick up from where you left off later.



The screenshot shows the AMD JupyterLab interface. At the top, the AMD logo is followed by "Welcome back!" and "Launch your GPU-powered Jupyter notebook". Below this is a text input field containing "mabhanda". Underneath is "Team: team-3904". A dropdown menu shows "Default (ROCm + vLLM)". A large blue button labeled "Launch Notebook" with a help icon is present. Below it, a light blue bar shows "3hrs 46min in 24hrs". At the bottom, there are two buttons: an orange "Restart Session" button and a red "Turn-off Session" button, both with help icons. A "Sign out" link is at the very bottom. Two blue arrows point from the text boxes on the left to the "Restart Session" and "Turn-off Session" buttons respectively.

AMD

Welcome back!

Launch your GPU-powered Jupyter notebook

mabhanda

Team: team-3904

Default (ROCm + vLLM)

Launch Notebook

3hrs 46min in 24hrs

Restart Session

Turn-off Session

[Sign out](#)

Persistence in a Running Session

Files in shared persist

Processes persist in a running session, when you reconnect.

The screenshot shows the JupyterLab web interface in a browser. The top bar indicates the session is titled 'JupyterLab' and the URL is 'notebooks.amd.com/jupyter-hack-team-3904-260606162157-a1'. The left sidebar shows a file browser for the '/ shared /' directory, containing two folders: 'plum' and 'unsloth', both modified '1h ago'. The main area displays a terminal window titled 'Terminal 6' with the prompt 'root@jupy'. The terminal output shows five 'hello world' messages, followed by a traceback for a 'KeyboardInterrupt' error. The status bar at the bottom shows 'Simple' mode, 6 CPU units, and 0 memory units.

Name	Modified
plum	1h ago
unsloth	1h ago

```
hello world
hello world
hello world
hello world
hello world

^CTraceback (most recent call last):
  File "<stdin>", line 2, in <module>
KeyboardInterrupt
hello world
>>> 
```

On Relaunching your Pod after it has Expired or was Shutdown

- What is in your persistent store is still available
- You need to re-install any software (pip install ..)
- Restart any processes you want (example LLM inference service)

Reminder: Four (4) Hours of Compute Resources in 24 hours

GPUs are an expensive resource and in limited supply. To ensure fair access to resources, we will time slice them by 4 hours in a 24-hour period, on a first come first serve basis per team.

NO GPUs? Try again in a little while!

Plan your work. Co-ordinate with your team.

Remember-2: Your Storage Disappears After the Hackathon

Each team will be granted 25 GB of storage for the period of the hackathon. Save datasets that you might download for training here. Perhaps a LORA model or two, some logs, or performance data that you might use in your presentations.

But it all disappears on June 15, 12 midnight. So, remember to download whatever you want to save to your laptop or other work environment, including your code. (`tar cvf something ...`)

Discord and Teams Support

- **Community/Expert Connect**

Stay connected with AMD experts for the duration of the Hackathon to get answers for your technical queries. Install Discord application on your **personal device**.

Sign-up for AMD Discord, Register using your personal email id & verify it.

Click on + to Join a Server

AMD TCS Hackathon specific: <https://discord.gg/tvVU2dhSB>

ADP (AMD Developer's Program) community :

<https://discord.com/invite/CxfXGhdp3J>

- Teams Kick-off
- Teams Office hours

AMD AI ACADEMY

 AI Academy

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Learning Paths



AI Foundations

Build a strong foundation in artificial intelligence, machine learning, and generative AI concepts.

 8 Courses

 15 Hour(s)

[View Pathway](#)



Build AI

Learn how to build intelligent AI-powered applications, agents, and multimodal workflows.

 5 Courses

 8 Hour(s)

[View Pathway](#)



Deploy & Optimize AI

Deploy, scale, and optimize AI systems for production using AMD GPUs, modern inference frameworks.

 6 Courses

 7 Hour(s)

[View Pathway](#)



Academic Developers

Learn how to empower teaching and research with AI through hands-on tutorials and modern

 6 Courses

 13 Hour(s)

[View Pathway](#)

[View All](#)

Learning Paths

Learning Paths

View All >

 AI Foundations Build a strong foundation in artificial intelligence, machine learning, and generative AI concepts while learning how to run and experiment with AI models on AMD platforms. 8 Courses 15 Hour(s) View Pathway →	 Build AI Learn how to build intelligent AI-powered applications, agents, and multimodal workflows using modern frameworks, open-source models, and scalable AI architectures. 5 Courses 8 Hour(s) View Pathway →	 Deploy & Optimize AI Deploy, scale, and optimize AI systems for production using AMD GPUs, modern inference frameworks, Kubernetes, and GPU performance optimization techniques. 6 Courses 7 Hour(s) View Pathway →	 Academic Developers Learn how to empower teaching and research with AI through hands-on tutorials and modern techniques that enhance coursework, support students, and advance research. 6 Courses 12 Hour(s) View Pathway →
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Recommended Courses

View All >

 Diffusion Models: From Prompts To Images And Video This four-part workshop demystifies diffusion models and shows how to turn text and images into high-quality videos and short videos. You'll learn the core components (tokenizer and CLIP text encoder, VAE latent space, UNet decoder, an...) 1 Hour(s)	 AI On AMD AI on AMD is an introductory course that teaches you how to run generative AI across AMD platforms, whether you have a Ryzen laptop, a Radeon GPU card, or a browser to access AMD Developer Cloud. You'll learn with by learning the high... 1 Hour(s)	 Finetuning Your Own R1 Reasoning Model On Unsloth In this hands-on workshop you'll learn how to convert any base LLM into a reasoning model that generates explicit chain-of-thoughts explanations. Using Group Relative Policy Optimization (GRPO) with Unsloth, we'll demonstrate a... 1 Hour(s)	 Introduction To RAG Introduces RAG techniques to enhance LLM capabilities and mitigate hallucinations, covering foundational RAG concepts, external knowledge sources, and retrieval-generator workflow. 2 Hour(s)
---	---	--	--

My Training

View All >

 AI Agents 101: Building AI Agents With MCP And Open-Source Inference Agents Progress 23%	 AI Agents 201: Design To Deployment: A Hands-On Guide To Multi-Agent Systems Agents Progress 20%	 GPU Optimization For LLM Inference Inference Progress 12%	 Hugging Face On AMD AI Fundamentals Progress 30%
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Resources

- [How to Get Started on the AMD Developer Cloud](#)
- [AMD MI300X server review 8x GPUs | Llama 405b model tested](#)
- [AMD Instinct™ MI300 series microarchitecture](#)
- [ROCm Software](#)
- [training-at-scale-white-paper.pdf](#)



Submission

- **Deadline – June 14 12 Midnight IST**
- Portal
- Components
 - Presentation
 - Demo Video
 - Code – GitHub link or code tar ball or WinZip file

Presentation (1-5 slides)

BASIC Information

- Project Title
- Short Description (what, why, how)
- Team name
- Team Members
 - name, email, role (research, coding, business case definition, ..)

Presentation (1-5 slides)

Problem & Context

- Problem Statement
- Current solutions, gap, what makes it important/difficult
- Target users/stakeholders/market size
- Mapped hackathon challenge

Presentation (1-5 slides)

Solution Overview

- Solution architecture / workflow diagram
- AI approach (RAG, fine-tuning, agents)
- Key technologies (AI frameworks, software packages, models)
- Datasets
- What was built as part of hackathon

Presentation (1-5 slides)

Details: Performance, Scale, Time

- Models
- Fine-tuning – datasets, time, epochs, accuracy, size .
- Throughput, end-to-end latency, concurrency, if pipeline
- GPU usage – memory, compute

Presentation (1-5 slides)

Summary

- Expected impact, value
 - (improves productivity, saves time, energy ..)
- Innovation/key differentiator Models
- Future work
- Demo video link
- Code link

Evaluation Criteria

Criteria	Description	Weightage
Problem Definition & Relevance	Clarity of problem, relevance to real-world / enterprise scenarios	10%
Technical Implementation	Correct and effective application of AI/ML techniques, working solution	40%
Learnings and Future work	Impact, usefulness, scalability, applicability	20%
Innovation & Creativity	Novelty, differentiation, unique approach	15%
Presentation & Demo Quality	Clarity of explanation, demo flow, storytelling	15%

Live Support Page

How to connect to Discord:

Discord Link:

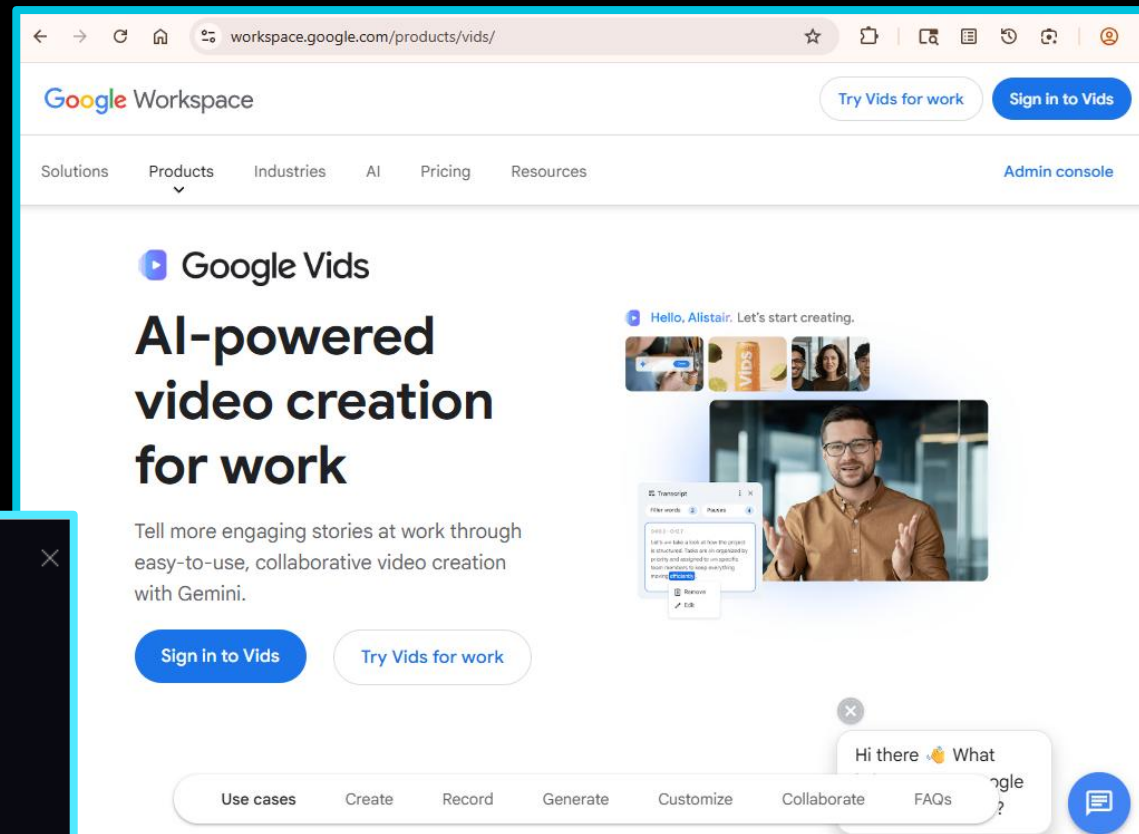
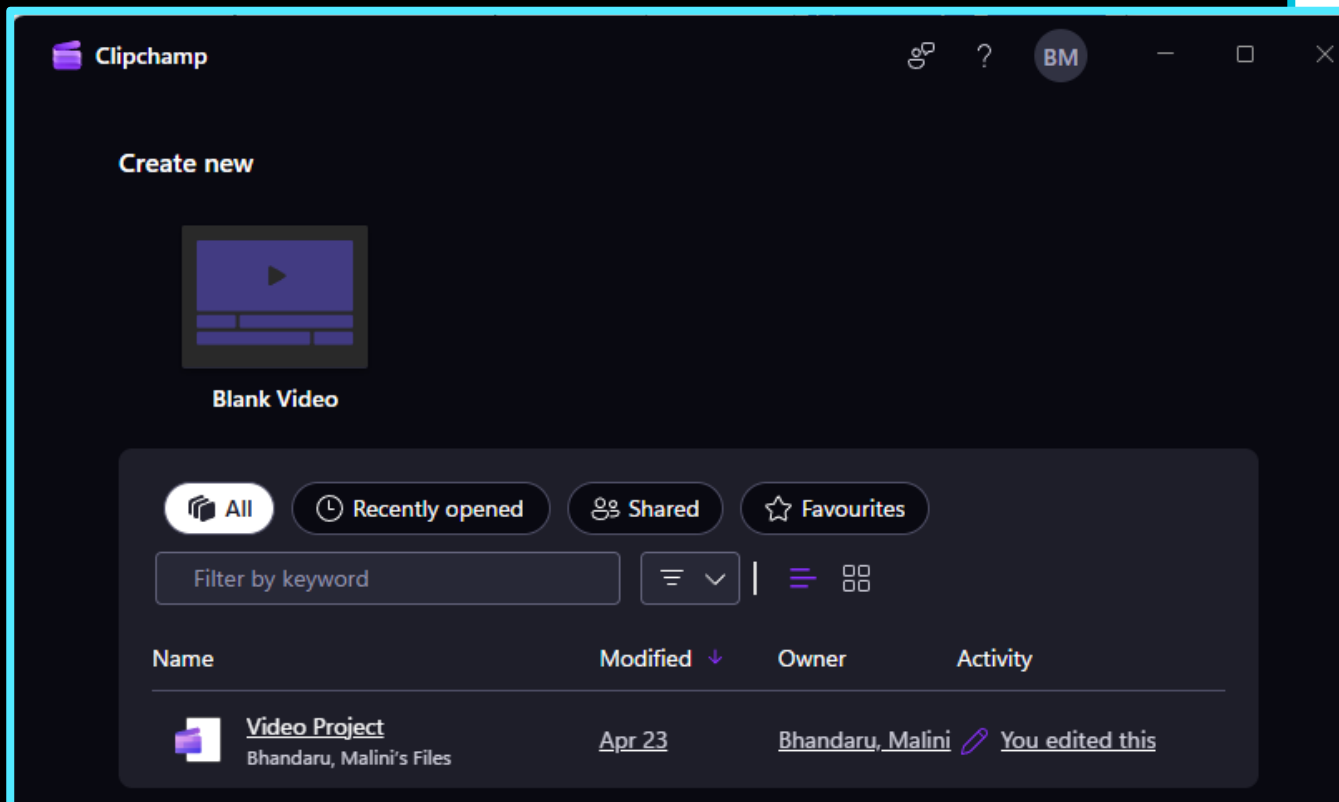
Reaching out to Discord Team members for Support:

Mentor Support: During the Hackathon, mentors will be available to provide constructive feedback and guidance. Engage with them by using the “Calling for Help” feature on your team’s page or by posting in the **#ineedhelp** channel on Discord and tagging **@Mentor**. Post-Kick-Off, an email will be sent to you with a document containing scheduling links for 1:1 mentor calls (not obligatory).

Demo Recording

Microsoft Clip Champ

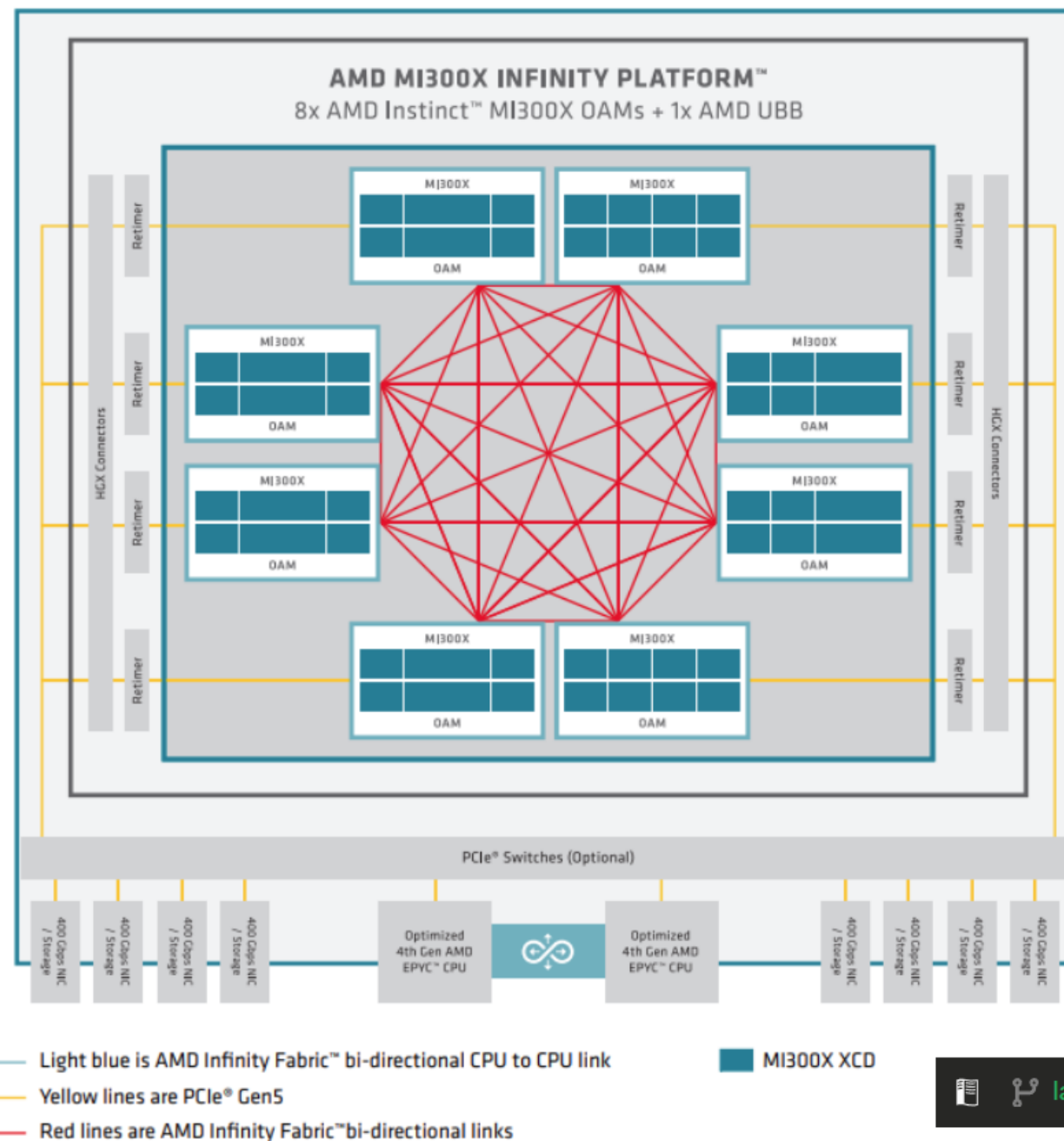
Google Vids: <https://workspace.google.com/products/vids/>



Have none of the above? Take screen shots of your demo and stitch into a presentation.

MI300 Node Level Architecture

Fully connected GPUs



Only Registered Participants

The screenshot shows a web browser window with the title "Dashboard - AMD AI Notebooks - [InPrivate]". The address bar shows the URL "https://notebooks.amd.com/tcs-hackathon". The page features the AMD AI Notebooks logo and the text "AMD AI Developer Program Portal". A user profile "mabhanda" is visible in the top right corner. The main content area displays a "Welcome back!" message with the instruction "Launch your GPU-powered Jupyter notebook". Below this, the username "mabhanda" is shown in a light blue box. A red error message states: "Your email is not registered for the hackathon. Please contact admins." A "Sign out" link is located at the bottom of the central card. The footer contains links for "Privacy Notice", "Cookie Policy", "Cookie Settings", "Terms of Use", and "Trademarks", along with the copyright notice "© 2026 Advanced Micro Devices, Inc. All rights reserved."

Dashboard - AMD AI Notebooks - [InPrivate]

https://notebooks.amd.com/tcs-hackathon

AMD AI Notebooks

AMD AI Developer Program Portal

mabhanda

AMD

Welcome back!

Launch your GPU-powered Jupyter notebook

mabhanda

Your email is not registered for the hackathon. Please contact admins.

[Sign out](#)

[Privacy Notice](#) [Cookie Policy](#) [Cookie Settings](#) [Terms of Use](#) [Trademarks](#)

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